

## 🔍 Kafka vs RabbitMQ (Complete End-to-End Comparison)

### System Overview

|                   |                                           |                                |
|-------------------|-------------------------------------------|--------------------------------|
| Category          | Kafka                                     | RabbitMQ                       |
| System Type       | Distributed Event Streaming Platform      | Message Broker / Message Queue |
| Core Model        | Distributed Commit Log                    | Message Queue with Routing     |
| Primary Use       | Event streaming, analytics, microservices | Task queues, async processing  |
| Message Retention | Time/size based                           | Until consumed/acked           |
| Replay Messages   | Yes                                       | No                             |

### Core Terminology

|                   |                   |                     |
|-------------------|-------------------|---------------------|
| Concept           | Kafka             | RabbitMQ            |
| Message container | Topic             | Queue               |
| Routing Layer     | Topic + Partition | Exchange            |
| Message Unit      | Record/Event      | Message             |
| Consumer Group    | Native            | Competing consumers |
| Partition         | Yes               | No                  |

### Message Distribution

|                |                          |                    |
|----------------|--------------------------|--------------------|
| Feature        | Kafka                    | RabbitMQ           |
| Load Balancing | Partition based          | Round robin        |
| Broadcast      | Multiple consumer groups | Fanout exchange    |
| Parallelism    | Partition level          | Multiple consumers |

### Consumption Model

|         |       |          |
|---------|-------|----------|
| Feature | Kafka | RabbitMQ |
|---------|-------|----------|

|                 |                        |                   |
|-----------------|------------------------|-------------------|
| Message Removal | Not deleted after read | Removed after ack |
| Offset Tracking | Consumer managed       | Broker managed    |
| Replay          | Yes                    | No                |

### Fault Tolerance

|                         |                          |                 |
|-------------------------|--------------------------|-----------------|
| Feature                 | Kafka                    | RabbitMQ        |
| Replication             | Leader/Follower replicas | Mirrored queues |
| Broker Failure Handling | Leader election          | Queue failover  |
| Durability              | acks + replication       | Durable queues  |

### Spring Boot Development

|                   |                      |                                 |
|-------------------|----------------------|---------------------------------|
| Area              | Kafka                | RabbitMQ                        |
| Dependency        | spring-kafka         | spring-boot-starter-amqp        |
| Producer API      | KafkaTemplate.send() | RabbitTemplate.convertAndSend() |
| Consumer Listener | @KafkaListener       | @RabbitListener                 |
| Config Beans      | Topic Bean           | Queue/Exchange/Binding Beans    |

### Architecture Flow

|                 |                                                     |                                           |
|-----------------|-----------------------------------------------------|-------------------------------------------|
| Flow            | Kafka                                               | RabbitMQ                                  |
| Processing Flow | Producer → Topic →<br>Partition → Consumer<br>Group | Producer → Exchange →<br>Queue → Consumer |